

Teenage Engineering EP-133 K.O. II — Cheat Sheet

128 MB sampler and composer, successor to the PO-33 K.O.I. 999 sample slots, 4 groups × 12 pads, 96 PPQN sequencer, 6 send FX + 12 Punch-In FX 2.0.

Top Panel Layout

Section	Controls
Display	Custom segmented display (66 icons), step position, parameter values
Mode	SOUND, MAIN (home), TEMPO (BPM/ time sig)
Function	FX, SAMPLE, KEYS (chromatic), TIMING (note interval / swing)
Transport	RECORD, PLAY, ERASE
Navigation	SHIFT, – / +, KNOB X, KNOB Y, FADER
Pads	12 velocity- & pressure-sensitive pads (also numpad 0–9, dot, enter)
Groups	GROUP A–D (pattern select, undo, copy, paste)
Top jacks	LINE IN (3.5 mm stereo), LINE OUT (3.5 mm stereo), SYNC IN/OUT, MIDI IN/OUT (TRS Type A), USB-C, power switch, volume knob

Project / Scene / Group / Pattern Structure

Concept	Details
Projects	9 per device, 80,000 notes each
Groups	4 per project (A–D), each with 12 pads
Patterns	99 per group, 1–99 bars each
Scenes	99 per project — snapshot of 1 pattern from each group
Song positions	99 — chain scenes into a full arrangement (up to 9,801 bars)

Sampling Workflow

1. Press **[SAMPLE]** — all pads blink, ready to record.
2. Press – / + to select input source: MIC, IN (line mono/stereo), RSP (resample mono/stereo), L.IN (left), R.IN (right), USB (mono/stereo).
3. **KNOB X** = input level, **KNOB Y** = threshold (auto-trigger when exceeded).
4. Hold a **pad** to start recording; release to stop.
5. Press the pad to audition. Press **[SAMPLE]** again or **[MAIN]** to exit.

Hands-free sampling: Hold **SHIFT** + press a pad to latch — K.O. II holds it for you. Press **PLAY** to record the pattern length. Press **[SAMPLE]** to stop.

Resample: Set source to RSP, latch a pad on an empty group, play to bounce your beat into a single sample.

Chop (SHIFT + SAMPLE): Auto-chop (intelligent or equal-length division) or live chop (tap pads to set slice points). Refine start/end with **KNOB X** / **KNOB Y**.

Sound Edit Pages (SHIFT + SOUND)

Page	KNOB X	KNOB Y
Play Mode	Oneshot / Key (poly) / Legato (mono)	Pan
Trim	Start point	Length / end point
Envelope	Attack	Release
Time Stretch	Mode (BPM / BAR)	Original BPM or bar division
MIDI	MIDI channel	MIDI root note
Mute Group	Select pads to group (cut each other off)	—

Hold **SHIFT + SOUND** for 2 seconds to save edits.

Key Shortcuts & Shift Combos

Action	Result
SOUND	Enter Sound mode (browse/assign samples)
SHIFT + SOUND	Sound Edit (6 pages)
MAIN	Home — sequencer, scenes, patterns
SHIFT + MAIN	Commit scene (duplicate & save, move to next)
TEMPO	Set BPM, metronome, time signature
SHIFT + TEMPO	Loop mode (OB-4 style, length with KNOB Y, slide with KNOB X)
FX	Select send FX for current group
SHIFT + FX	Master compressor / sidechain settings
SAMPLE	Enter sampling mode
SHIFT + SAMPLE	Chop mode
KEYS	Chromatic keyboard mode for selected pad
TIMING	Note interval, swing, quantize / free time
SHIFT + TIMING	Timing correct (per-pad quantize)
SHIFT + ERASE	System settings
SHIFT + GROUP A	Find next empty pattern
SHIFT + GROUP C	Copy (press once = bar, twice = pattern)
SHIFT + GROUP D	Paste
SHIFT + FADER	Reset fader position
SHIFT + PLAY	Play from start
SHIFT + - / + (Sound mode)	Jump sounds by 10
Hold TIMING + pad	Note repeat (pressure-sensitive)
Hold TIMING + SHIFT + pad	Latch note repeat

Hold RECORD + pad	Step-record a note
Hold RECORD + move FADER	Record fader automation
Hold FX + pads	Punch-In FX (pressure-sensitive, combinable)
Hold FX + group pad	Solo group

Fader Assignments

Hold **FADER** + press pad to assign:

Level (default) · Pan · FX Send · Pitch · Attack · Release · Filter Cutoff · Filter Resonance

Send FX (6 types)

Effect	KNOB X	KNOB Y
Delay	Length (time)	Feedback (repeats)
Reverb	Length (room size)	Color (filter dark/bright)
Distortion	Drive (overdrive)	Color (filter + resonance)
Chorus	Modulation (rate)	Feedback (prominence)
Filter	Cutoff (LPF / HPF)	Resonance
Compressor	Drive (input gain)	Speed (reaction time)

Punch-In FX 2.0

Hold **FX** + press pads for real-time performance effects. Pressure-sensitive and combinable (e.g. LPF + HPF = band-pass).

Pad	Effect	Pressure control
0	Pitch Randomiser	Decreasing pitch
ENTER	Granuliser	Grain Size
1	Beat Repeat	Beat Length
2	Tape Stop	Slow Down
3	Filter LFO	LFO Speed
4	Low-Pass Filter	Cutoff Frequency
5	High-Pass Filter	Cutoff Frequency
6	Send FX	FX Depth
7	Tremolo	Speed
8	Octave Down	Increasing Pitch
9	Decimator	Bit / Sample Reduction
. (dot)	Sample Swap	—

Note Intervals (TIMING + KNOB X)

1/1 · 1/2 · 1/4 · 1/8 · 1/8T · 1/16 · 1/16T · 1/32

Swing applies to 1/8 and 1/16 (KNOB Y).

Connectivity

- **LINE IN (3.5 mm)** — stereo input for instruments, phones, turntables (with RIAA preamp)
- **LINE OUT (3.5 mm)** — stereo output / headphones
- **SYNC IN / OUT** — 8th note, 16th note (Eurorack), or Sync 24 (vintage drum machines)
- **MIDI IN / OUT** — TRS Type A, 3.5 mm
- **USB-C** — MIDI, firmware, sample transfer (EP Sample Tool), power (5V / 1A)
- **Built-in mic** — mono
- **Built-in speaker** — mono

Specs at a Glance

Spec	Value
Sample memory	128 MB (early units: 64 MB)
Sample slots	999
Polyphony	16 mono / 12 stereo voices
Sampling rate	46.875 kHz / 16-bit (adjustable lo/mid/hi)
Internal signal chain	32-bit float, 24-bit ADC/DAC
Groups × pads	4 × 12
Patterns per group	99 (up to 99 bars)
Scenes per project	99
Projects	9
Sequencer	96 PPQN
Effects	6 send FX + 12 Punch-In FX + master compressor
Power	4× AAA batteries or USB-C
Dimensions	240 × 176 × 16 mm
Weight	620 g

Quick Tips

- Default sound categories: 1–99 kicks, 100–199 snares, 200–299 hats, 300–399 percussion, 400–499 bass, 500–599 melodic.
- **Commit often** (SHIFT + MAIN) — it duplicates the scene so you can evolve the arrangement without losing your place.
- Use **SHIFT + FADER** to reset the virtual fader and undo automation.
- Hold **SHIFT** while turning knobs or moving the fader for fine-grained adjustments.
- Set a pad's sample to 000 in Sound mode to create an empty MIDI-only pad that doesn't use a voice.
- **Sidechain**: SHIFT + FX → press + to access note-triggered sidechain. Select source sound, destination groups, length (KNOB X), and shape (KNOB Y).

- **Live input FX:** plug into LINE IN, press MAIN, turn up KNOB Y (FX send level) to process external audio through the selected effect.
- USB audio (class compliant) — use KO2 as an audio interface to your phone or computer.

Source: *teenage.engineering guides*, *gabrielroth.com/ko2 manual (firmware v2.0.5)*